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CSA - 1306

CO2 - ASSIGNMENT 2

TASK - 1 : Regular expression design for input validation

(i) $(0+1)^*01$

~~Sample input :-~~ 01
101
0101

Represents binary strings that end with suffix 01 over $\Sigma = \{0,1\}$

(ii) $(0+1)^*(11+00)(0+1)^*$

Binary strings containing substring 11 (or) 00 over $\Sigma = \{0,1\}$

(iii) $(a+b)^*ab(a+b)^*$

Strings over $\Sigma = \{a,b\}$ containing substring ~~conta~~ ab

(iv) $1^*(01^*01^*)^*$

Strings over $\Sigma = \{0,1\}$ having even number of zeroes.

(v) $a(a+b)^*b$

Strings over $\Sigma = \{a,b\}$ that start & end with a & b respectively.

TASK - 2 : Regular expression analysis & language representation

Given, $R = (0+1)^*101(0+1)^*$

(i) The Language represented by the regular expression is set of binary strings containing the substring "101".

(ii) Alphabet $\longrightarrow$ $\Sigma = \{0,1\}$

Substring $\longrightarrow$ 101

(iii)

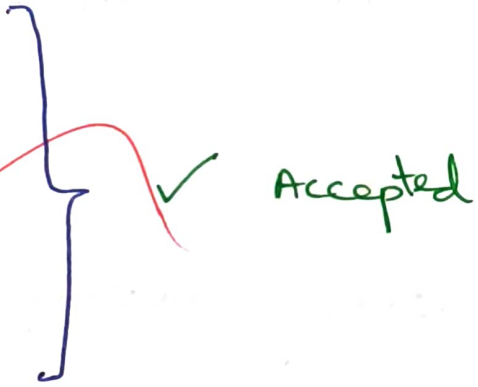
101

00101

10101

1110101

100101001



(iv)

0

1

100



(v) Kleene-star : Repetition of zero (or) more times

Union operations : Acts as logical choice/OR operator

Together, they contribute to the syntax of this regular expression for this Language.

TASK - 3 : Context-free grammar construction

(i) $L_1 = \{a^n b^n \mid n \geq 0\}$

$G = \{V, T, P, S\}$; where $V \rightarrow \{S\}$

$T \rightarrow \{a, b\}$

$P \rightarrow S \rightarrow aSb \mid \varepsilon$

$S \rightarrow S$

(ii) $L_2 = \{a^n b^m \mid n, m \geq 0\}$

$G = \{V, T, P, S\}$; where $V \rightarrow \{S, A, B\}$

$T \rightarrow \{a, b\}$

$P \rightarrow S \rightarrow AB$

$A \rightarrow aA \mid \varepsilon$

$B \rightarrow bB \mid \varepsilon$

$S \rightarrow S$

(iii) $G = \{V, T, P, S\}$; where $V \rightarrow \{S\}$

$T \rightarrow \{id, +\}$

$P \rightarrow S \rightarrow S + id$

$S \rightarrow id$

$S \rightarrow S$

(iv) Balanced parantheses

$G = \{V, T, P, S\}$; where,

V → {S}

T → {(,)}

P → S → (S)S

S → ε

S → S

(v) G = {V, T, P, S}

V → {S}

T → {0, 1}

P → S → 0S1S

S → 1S0S

S → ε

S → S

TASK - 4 : Derivation & parse tree construction

Given :- S → aSb | ∈

$L = \{a^n b^n \mid n \geq 0\}$

(i) V → {S}

T → {a, b}

P → S → aSb | ∈

S → S

(ii) $n=0 \Rightarrow S \Rightarrow \epsilon$

$n=1 \Rightarrow S \Rightarrow aSb \Rightarrow ab$

$n=2 \Rightarrow S \Rightarrow aSb \Rightarrow aaSbb \Rightarrow aabb$

$n=3 \Rightarrow S \Rightarrow aSb \Rightarrow aaSbb \Rightarrow aaaSbbb \Rightarrow aaabbb$

(iii) LMD :-

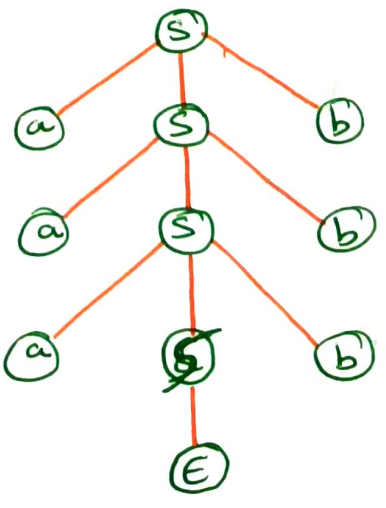
S
↓
aSb
↓
$aaSbb$
↓
$aaaSbbb$
↓
$aaa(\epsilon)bbb$
↓
$aaabbb$

(iv) RMD :-

S
↓
aSb
↓
$aaSbb$
↓
$aaaSbbb$
↓
$aaa(\epsilon)bbb$
↓
$aaabbb$

Since there is only 1 Non-terminal value at each step, LMD & RMD are the same.

(v) Parse tree :-



TASK - 5 :- Grammar design for Arathmetic expression

$G = \{V, T, P, S\}$

$V \longrightarrow \{E, T, F\}$

$T \longrightarrow \{a, b, c, +, *, (,)\}$

$P \longrightarrow$ $E \longrightarrow E+T$

$E \longrightarrow T$

$T \longrightarrow T*F$

$T \longrightarrow F$

$F \longrightarrow (E)$

$F \longrightarrow a|b|c$

$S \longrightarrow E$

Operation precedence :-

E (expression) : Addition

T (Term) : Multiplication

F (factor) : identifiers & parentheses

$E < T < F$

LMD :-

E

↓

E + T

↓

T + T

↓

F + T

↓

a + T

↓

a + T * F

↓

a + F * F

↓

a + b * F

↓

a + b * c



Parse tree :-

